

SOFTWARE ENGINEERING PROJECT REPORT-1

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Abstract— This document presents the complete set of deliverables prepared for the development of an educational content platform, structured according to IEEE formatting principles. The work consists of four major components: (1) well-defined user stories with their associated scenarios describing end-user interactions, (2) a Software Requirements Specification (SRS) document excluding system-level requirements and detailing functional, data, and interface specifications, (3) a project backlog outlining implementation tasks with priority levels and their rationales to guide development sequencing, and (4) a set of dummy or simulated graphical interfaces demonstrating the expected appearance and behavior of key frontend screens. Together, these components establish a unified foundation for understanding user needs, system behavior, development workflow, and planned UI layouts for both web and mobile environments.

Keywords—user stories, software requirement specification, backlog management, graphical interface simulation, frontend-backend integration, role-based systems.

I. USER STORY

Ayşe is a high school student preparing for the university entrance exam, and she uses an online learning application to improve her weaknesses in mathematics. After logging into the application, Ayşe sees information such as “Continue Where You Left Off,” “Your Ranking,” and “Reward Table” on the homepage. To choose the topic she wants to study, she navigates to the topic selection page from the bottom menu.

The system displays topic categories divided into TYT, AYT, and grade levels. If Ayşe selects TYT, the system shows the list of subtopics under the TYT category. If Ayşe selects “Basic Concepts” from this list, the system directs her to the material selection screen for that topic.

On this screen, the system offers Ayşe three options: Topic Explanation, Videos, and Tests.

If Ayşe selects Topic Explanation, the system displays the list of PDF documents related to that topic. When Ayşe chooses a PDF from the list, the system opens it in full screen, allowing her to scroll through the pages. If she closes the document, the system returns her to the material selection screen.

If Ayşe selects Videos, the system displays the video list for the Basic Concepts topic. When Ayşe taps on a video, the

system opens the video player and starts playback. If Ayşe watches the video until the end, the system detects the completion and updates her progress. If she closes the video halfway, the system returns to the previous screen without marking the video as completed.

If Ayşe selects Tests, the system shows the test list for that topic. When Ayşe starts a test, the system presents the questions one by one. If she completes the test, the system calculates her correct and incorrect answers, generates a score, and saves it to her progress data. If she leaves the test halfway, the system does not produce a result and returns her to the test list.

After finishing her study session, if Ayşe taps the profile icon in the bottom menu, the system displays options such as account information, user settings, app settings, and progress. If Ayşe opens the Progress section, the system lists all videos she has completed, tests she has solved, and PDF documents she has viewed. If she wants more detail, the system opens the corresponding progress card and provides additional information.

If Ayşe wants to quickly find a specific topic, she can tap the search icon. When she types the first letters of a keyword into the search bar, the system lists the matching topic suggestions. If Ayşe selects one of these suggestions, the system takes her directly to the material screen for that topic.

Throughout the application, whichever option Ayşe chooses, the system initiates the appropriate process; at every step, it checks her selections, records completion states, and navigates her to the correct next screen when necessary.

II. SOFTWARE REQUIREMENTS SPECIFICATION DOCUMENT

A. Introduction

The increasing demand for online educational systems requires platforms that securely handle user authentication, manage multimedia educational content, and provide reliable access across both web and mobile clients. To meet these requirements, the proposed system integrates a unified frontend (React Web + React Native Mobile), a backend service (Spring Boot), and cloud-based object storage to support efficient upload, retrieval, and streaming of learning materials.

This SRS document defines the functional specifications for the unified system, focusing on user management, role-based authorization, content upload and viewing functionalities, and frontend-backend interaction behaviors. Hardware, deployment constraints, and other system-level requirements are not included in this phase of documentation.

This document follows IEEE SRS structural conventions. Section II-A introduces the system context, Section II-B describes product perspective and functions, Section II-C details user classes and constraints, Section II-D provides functional and data requirements, Section II-E lists external interface specifications, and Section II-F outlines non-functional requirements. Phase-1 deliverables and acceptance criteria are presented at the end of this section.

B. Overall Description

1) Product Perspective

The system is designed using a three-tier architecture that integrates a React-based web application, an Expo (React Native) mobile application, and a Spring Boot backend. The frontend provides role-based interfaces, including file upload panels for Uploaders and viewing interfaces for all users, supporting both PDF documents and video playback. The backend follows a layered REST architecture composed of Controllers, Services, and Repositories, and it handles authentication using JWT, file upload processing through multipart or pre-signed URLs, centralized validation, and metadata storage. At the data layer, PostgreSQL is used to manage user information, role-based authorization tables, and content metadata stored in the users, roles, user_roles, and contents tables.

2) Product Functions

The system offers a variety of core functions to support secure and efficient educational content delivery. Users can register and authenticate using email and password credentials, after which JWTs are issued and validated for continued access. Role-based authorization ensures that Uploaders can upload materials while Viewers can only consume content. The platform supports uploading files directly to the backend or through pre-signed URLs, retrieving and listing available content, and delivering videos with HTTP Range streaming for efficient playback. PDF files are displayed securely, and both the web and mobile clients communicate with the backend through RESTful interfaces.

3) User Classes

Two primary user roles interact with the system: Uploaders and Viewers. Uploaders are granted permissions to upload PDF or video content and are provided with access to interfaces and endpoints related to content management. Viewers, on the other hand, cannot upload files but have full access to viewing and streaming functionality. Their interface and endpoint access are intentionally limited to ensure a streamlined, read-only interaction experience.

4) Operating Environment

The system operates across multiple environments and platforms. The web interface runs on React 18+ and is compatible with modern browsers such as Chrome, Edge, and Safari. The mobile application is built using Expo SDK 54 and supports both Android and iOS devices. The backend relies on Spring Boot 3.x with Java 21 and communicates with a PostgreSQL 15+ database. Production environments require secure HTTPS communication, while development environments may rely on HTTP or local networking setups.

5) Design and Implementation Constraints

Several constraints shape the design and implementation of the system. All user passwords must be hashed using BCrypt, and JWTs must include user role information to support proper authorization. Web access is restricted through CORS, allowing only approved domains. Strict adherence to REST conventions is required for consistency, and production deployments must enforce HTTPS to ensure secure communication.

6) Assumptions and Dependencies

The system design assumes the availability of an S3-compatible object storage service, such as Amazon S3 or MinIO, for hosting uploaded content. Both the web and mobile applications are expected to use the same backend API base URL provided through environment-specific configuration. Additionally, offline-first functionality is not required within the scope of this phase, simplifying storage and synchronization requirements.

III. PROJECT BACKLOG DOCUMENT WITH TASK PRIORITIES AND THEIR RATIONALES

This section presents the backlog of the project organized into EPICs. Each task includes its identifier, priority level, and a concise rationale. The structure is optimized for IEEE conference formatting where table-based layouts may collapse due to column width constraints..

A. EPIC 1 — Authentication (Frontend + Backend)

Backend Tasks

BE-01 — Implement /auth/register (High): Essential entry point for onboarding new users; system cannot function without registration.

BE-02 — Implement /auth/login with JWT generation (High): Required for secure session creation and all protected actions.

BE-03 — Add BCrypt password hashing (High): Ensures safe password storage and prevents credential leaks.

BE-04 — Implement /auth/me (High): Frontend needs this endpoint to retrieve user identity and roles.

Frontend Tasks

FE-01 — Registration screen UI (High): First touchpoint for new users; must be available early.

FE-02 — Login screen UI (High): Required to access protected parts of the system.

FE-04 — Connect login/register APIs (High): Enables real authentication instead of static screens.

FE-06 — Implement secure token storage (High): Prevents unauthorized reuse of user sessions.

B. EPIC 2 — Role-Based Access Control (RBAC)

Backend Tasks

BE-10 — Add roles claim to JWT (High): Necessary for differentiating Uploader/Viewer access.

BE-11 — Add role-based protection to upload/delete endpoints (High): Prevents Viewers from modifying resources.

Frontend Tasks

FE-09 — Decode roles from JWT (High): Required for adaptive UI rendering.

FE-11 — Hide upload UI for Viewers (High): Prevents unauthorized or misleading interactions.

C. EPIC 3 — Content Upload (Uploader Only)

Backend Tasks

BE-20 — Implement multipart upload handler (High): Core functionality enabling Uploaders to provide content.

BE-21 — Validate file type and size (High): Protects system from corrupted or oversized files.

BE-22 — Generate pre-signed URLs for large uploads (Medium): Reduces backend load and supports scalability.

Frontend Tasks

FE-14 — Upload screen UI (Web + Mobile) (High): Enables Uploaders to initiate file submission.

FE-17 — Connect upload UI to backend API (High): Creates a functional file submission workflow.

FE-19 — Add upload progress indicator (Medium): Improves UX for large file uploads.

D. EPIC 4 — Content Listing & Viewing (All Users)

Backend Tasks

BE-30 — Implement /contents listing endpoint (High): Main feed for accessing available materials.

BE-31 — Provide pre-signed URLs for content access (High): Ensures secure delivery of files.

Frontend Tasks

FE-21 — Content list UI (High): Central navigation interface for all users.

FE-25 — Content detail screen (High): Displays metadata before viewing.

FE-27 — PDF viewer integration (High): Required for document-based content.

FE-26 — Video player integration (High): Enables secure streaming.

E. EPIC 5 — User Profile & Session Management

Backend Task

BE-32 — Improve session validation (Medium): Enhances reliability and reduces token issues.

Frontend Tasks

FE-30 — Profile screen UI (Medium): Displays user data and role.

FE-31 — Connect /auth/me to profile screen (Medium): Replaces placeholders with live data.

FE-32 — Logout functionality (High): Required for secure session termination.

FE-34 — Axios interceptor for JWT (High): Ensures all API calls include credentials.

F. EPIC 6 — Integration Layer (Frontend + Backend)

Backend Tasks

BE-40 — CORS configuration (High): Required for the web client to communicate with the API.

BE-41 — Unified error response format (Medium): Simplifies frontend error handling.

Frontend Tasks

FE-42 — Centralized Axios API client (High): Avoids repeated configuration across the codebase.

FE-43 — Global error handler (High): Provides consistent user feedback on network/API issues.

FE-45 — Environment-based BaseURL switching (High): Prevents environment-driven connection failures.

G. EPIC 7 — Deployment Preparation

Backend Task

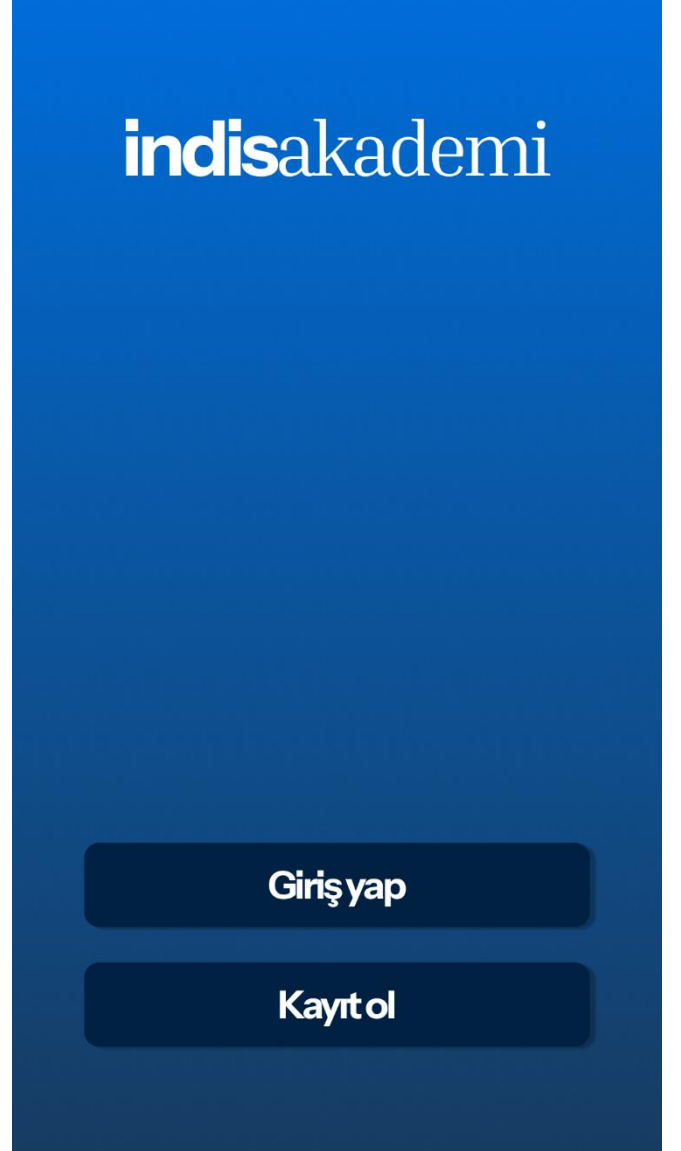
BE-50 — API health check endpoint (High): Allows deployment scripts and clients to verify backend status.

Frontend Tasks

FE-48 — Configure environment variables (High): Protects sensitive data and supports multi-environment builds.

FE-49 — Production build for Web + Expo EAS (High): Required for testing and deployment.

IV. DUMMY/SIMULATED GRAPHICAL INTERFACE



Giriş yap

Kullanıcı adı veya e-posta

Şifre



[Şifremi unuttum](#)

Giriş yap

[Hesabın yok mu? Hemen kaydol!](#)

indisakademi

Kayıt ol

İsim-Soyisim

Kullanıcı adı

E-posta

Telefon numarası

Şifre



Kayıt ol

[Hesabın var mı? Giriş yap!](#)

indisakademi



Aktivasyon

E-postana gelen 6 haneli doğrulama kodunu gir

Doğrula

Kodu yeniden gönder

indisakademi

Paketler

Paket içeriğini gör

Paket içeriğini gör

Paket içeriğini gör

Planını seç, öğrenmeye başla!



Hoşgeldiniz

Robb Stark

Kaldığınız Yer

Türevin Tanım Kuralları

Sıralamanız

36/821

Ödül Tablosu

Cumhuriyet Altını



fonksiyon



birim fonksiyon
doğrusal fonksiyon
bileşke fonksiyon
parçalı fonksiyon
ters fonksiyon
içine fonksiyon
örten fonksiyon
birebir fonksiyon



Profil

Robb Stark

Hesap Bilgileri

Kullanıcı Ayarları

Uygulama Ayarları

İlerleme

Sıralama



Materyal Seçimi

Robb Stark

Videolar

Konu Anlatımlar

Testler

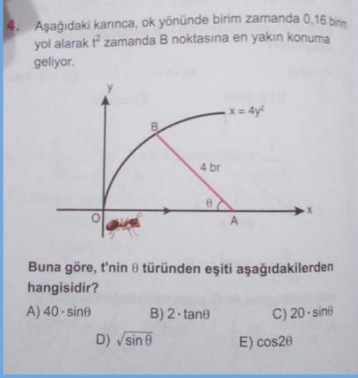


ChatBlog

Robb Stark

Ayşe Fatma

θ Açısını nasıl kullanacağız anlamadım. Yardımcı olabilir misiniz?



Mehmet Selim

$d^2 = (4 - 0,16t)^2 + (1 - 0)^2$ formülüyle yapabilirsin

Ayşe Fatma

Sayende soruyu çözdüm. Çok teşekkürler!

Dosya Ekle



Konu Seçimi

Robb Stark

TYT

AYT

9. Sınıf

10. Sınıf

11. Sınıf

12. Sınıf



Konu Seçimi

Robb Stark

TYT

Temel Kavramlar

Sayı Basamakları

Bölme ve Bölünebilme

EBOB – EKOK

Rasyonel Sayılar

Basit Eşitsizlikler

Mutlak Değer

Üslü Sayılar

Köklü Sayılar

Çarpanlara Ayırma

Oran Orantı

Denklem Çözme



Temel Kavramlar

Video 1



Video 2



Video 3



Temel Kavramlar

Test1

Test2

Test3

Test4

Test5



Temel Kavramlar

Pdf1

Pdf2

Pdf3



Exclusive

- Farklı seviyelerde çeşitli sorular
- Detaylı konu anlatımları
- Açıklamalı soru çözümleri
- Ücretsiz sınav planlayıcısı
- Test analizleri
- Deneme sınavları
- Ödüller

Paketi al



Premium

- Farklı seviyelerde çeşitli sorular
- Detaylı konu anlatımları
- Açıklamalı soru çözümleri
- Ücretsiz sınav planlayıcısı
- Test analizleri
- Deneme sınavları
- Ödüller

Paketi al



Free

- Farklı seviyelerde sorular
- Konu anlatımı
- Açıklamalı soru çözümü
- Ücretsiz sınav planlayıcısı
- Test analizi

Paketi al



indisakademi

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Temel Kavramlar

Pdf1

Pdf2

Pdf3

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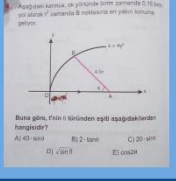
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Yardımcı olabilir misiniz?



Buna göre, theta i birimden eşit sapıklardan hangisidir?
A) 45-grad B) 2-grad C) 20-grad
D) 1-grad E) 30-grad

Mehmet Selim
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Temel Kavramlar

Video1

Video2

Video3

Video4

Döküman Ekle

Materyal Seçimi

Konu Seçimi

Dosya Ekle

Temel Kavramlar

Test1

Test2

Test3

Test4